# StoryPoints AI Internship Program

## Week 2 Assignment — Data Orchestration, Validation & Monitoring in GCP Composer

### 🎯 Learning Goals

- Understand Cloud Composer (Airflow in GCP)
- Orchestrate ingestion, transform, and load tasks as a DAG
- Add data validation checks before loading
- Track metadata (rows processed, validation status)
- Implement basic alerts when pipelines fail

### 📌 Scenario

In Week 1, you built ETL scripts to fetch, transform, and store data. Now your manager wants you to operationalize this pipeline in GCP Composer (Airflow). The goal is to orchestrate tasks, validate data quality, track metadata, and set up monitoring and alerts.

### 🛠️ Tasks

1. Set up Cloud Composer in GCP (see Quickstart docs).

2. Build a DAG orchestrating: ingest_clickstream → ingest_transactions → ingest_currency_api → transform → load_to_gcs.

3. Add validation before load_to_gcs (null checks, positive amounts, valid currency codes).

4. Track metadata: rows ingested, transformed, loaded, validation status. Store in BigQuery or CSV in GCS.

5. Monitoring & Alerts: configure Airflow alerts (email/Slack) for failures. Log alerts separately in alerts.log.

6. Documentation: update README with DAG screenshot, validation logic, metadata tracking approach.

## 🧑‍💻 Coding Standards

❌ No hardcoded values in DAGs or scripts.

✅ Use Airflow Variables, GCP Secret Manager, or environment variables for configs and secrets.

All scripts must have docstrings and clear inline comments.

Secrets like API keys must be externalized, not in code.

Follow Week 1 pro tips (chunked ingestion, retries, UTC timestamps, partitioning, structured JSON logging, idempotence).

## 📦 Deliverables

/week2_orchestration/dags/etl_week2_dag.py

/week2_orchestration/validation/validation.py

/week2_orchestration/metadata/run_log.csv (or BigQuery table)

Updated README.md with DAG flow and validation details

## 📊 Evaluation Criteria

| Aspect | Weight |
|---|---|
| Airflow DAG Correctness & Readability | 25% |
| Data Validation Rules | 25% |
| Metadata Tracking | 20% |
| Monitoring & Alerts | 20% |
| Documentation | 10% |

## 💡 Pro Tips

Use PythonOperator to wrap Week 1 scripts.

Set TriggerRule=all_success to enforce dependencies.

Design tasks to be idempotent — reruns should not duplicate data.

Use Composer logs and Airflow UI to debug failures.

Store metadata in BigQuery for better visibility.